

Deep Learning Introduction

Deep Learning is a subset of Machine Learning based on Artificial Neural Networks.

Neural Network Components

- Input Layer
- Hidden Layer
- Output Layer

Activation Functions

- ReLU
- Sigmoid
- Tanh

Convolutional Neural Networks (CNN)

CNNs are mainly used for image classification.

Recurrent Neural Networks (RNN)

RNNs process sequential data such as text.

Transformers

Transformers use the Attention mechanism and are widely used in Natural Language Processing.

Applications

- Image Recognition
- Speech Recognition
- Chatbots
- Machine Translation